ECN 712 Fall 2025
Professor EE Schlee
Problem Set VIII, Due 6 November

1. A monopoly produces and sells a single output under constant returns to scale. Its unit cost of production, however, depends on how much it invests in technology before production; in particular, if the firm spends I dollars, then its *average* cost of production is $c(I) > 0$ where $c(I) \geq \bar{c} > 0$ for all $I \geq 0$ for some constant $\bar{c}$, and the function $c(\cdot)$ is *strictly decreasing*. The inverse demand $P(\cdot)$ satisfies $P(0) > c(0)$ and $P(q) = 0$ if and only if $q \geq \bar{q}$ for some $\bar{q} > 0$. The function $P(\cdot)$ is continuous and *strictly decreasing* on $[0, \bar{q})$.[1]

   (a) Write down the firm's problem who chooses (q, I) to maximize profit.

   (b) Write down the problem of a planner who chooses (q, I) to maximize total surplus.

   (c) Fix $q \geq 0$ for both the planner and firm and consider the choice of I. Let I_1 solve the planner's problem and I_0 solve the firm's problem at q. Does it follow that $I_1 \geq I_0$? Why or why not?

   (d) Let (q_0, I_0) be any solution to the firm's problem in part (a) and (q_1, I_1) any solution to the planner's problem in part (b).

      i. Prove that $q_1 \geq q_0$.

      ii. Prove that there is a solution (q_1', I_1') to the planner's problem with $(q_1', I_1') \geq (q_0, I_0)$.

      iii. Suppose in addition that c is strictly convex. Prove that $(q_1, I_1) \geq (q_0, I_0)$.

2. MWG, page 262, Exercise 8.B.3. Assume that preferences are quasilinear in some good other than the one being auctioned and that bidder valuations are positive. You only need to show that $(b_1, ..., b_I) = (v_1, ..., v_I)$ is a dominant strategy equilibrium. (As is my usual advice, if you have trouble with $I > 2$ bidders, start with $I = 2$.

3. Consider a first-price sealed-bid auction of a single indivisible object to two bidders. Bidder i's valuation is v_i where $v_1 > v_2 > 0$. Each player i submits a bid b_i, a nonnegative real number, which is put into an envelope. The envelopes are opened, and the bidder with the highest bid gets the object and pays the auctioneer the highest bid. If the bids are the same the winner is determined by the toss of a fair coin. (Bidders are risk neutral.)[2]

   (a) Identify any strictly dominated strategies for each player.

   (b) Identify any weakly dominated strategies for each player.

   (c) Identify any Nash equilibria of the game.

4. MWG, Exercise 8.D.4.

[1] This question is a sharpening of MWG 12.B.9.
[2] This problem is a variant of MWG 8.D.3.